Vidhi Khakhar

Website | LinkedIn | GitHub

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Introduction

Second-year Computer Science and Engineering student at Delft University of Technology, with experience in Java, Scala, and Python and coursework in Machine Learning, Big Data Processing, and Software Engineering. Interested in agentic AI and large-scale data systems, particularly the engineering and evaluation methods needed to make AI systems reliable and secure in production. Motivated by challenging problems at the intersection of machine learning, distributed systems, and software engineering, and eager to apply rigorous research methods and systems thinking to building reliable software systems for real-world use.

Education

2025 - 2028 (Expected)
Delft University of Technology, *BSc Computer Science and Engineering*
Relevant Coursework:
Machine Learning (Python), Big Data Processing (Scala, Apache Spark, Apache Flink), Algorithms and Data Structures, Probability and Statistics, Linear Algebra, Calculus, Reasoning and Logic, Object-Oriented Programming (Java)

2009 - 2024
International School of The Hague, *International Baccalaureate (IBDP) Diploma*
Received a Bilingual Diploma, English and Dutch
- Higher Level: Computer Science, Mathematics: Analysis & Approaches, Business Management
- Standard Level: Physics, English A Language & Literature, Dutch A Language and Literature

2022 - 2023
Harvard University, *Harvard Business School - Summer Programme*
Sustainable Product Design and the Innovation Ecosystem - *Final Grade: A. 99/100.*
Investigated how sustainable product design drives competitive advantage through technology; built a business plan on industrial AR/VR use cases.

Experience

AI Intern - Vattenfall, Aug 2025 - Nov 2025
- Analyzed developer workflows and information bottlenecks on a regulated energy-trading platform
- Researched RAG architectures and Azure AI services, then built a cloud-based proof of concept (Microsoft Azure, AI Foundry, AI Search, Blob Storage, Copilot Studio, Power Automate) integrating vector-indexed source code with Azure DevOps data
- Worked with Data Privacy and Security Officers to address governance requirements and secure approval for use of the enterprise AI platform
- Evaluated the system on representative workflows, documented its design and limitations, and identified opportunities to improve developer efficiency and project insight
- Delivered a Proof of Concept, along with Design, Build, and Test documentation

Projects and Mentorship

Teaching Assistant and Mentor - Delft University of Technology, Aug 2026 - Present

- As a mentor, guided a cohort of 23 first-year students through their transition to university via weekly academic and support meetings; advised on study strategies and navigating university.
- Collaborate with lecturers across multiple Computer Science courses to support students in developing rigorous technical reasoning, debugging strategies, and problem-solving skills
- Teach and guide students during lab sessions by identifying misconceptions, asking targeted questions, and translating complex concepts into clear explanations
- Evaluate assignments and examinations against academic criteria while maintaining consistency across large student cohorts
- Teaching Assistant for:
    - Introduction to Programming: Object-Oriented Programming in Java
    - Computer Organization: Assembly x86-64, Logic Gates, Binary Representation & Operations
    - Reasoning and Logic: Propositional & Predicate Logic, Mathematical Proofs, and Puzzles

NetNote Software – **Collaborative Note-Taking App** *(Java 21, Spring Boot 3, JavaFX)*

- Built a collaborative Markdown note-taking system in a four-person Agile team, using Spring Boot and JavaFX with REST APIs and WebSockets for client-server communication and real-time synchronization.
- Implemented real-time Markdown rendering, search, collection-based organization, embedded media, and multilingual support.
- Contributed through weekly sprint planning, GitLab-based version control, merge requests, and peer code reviews.

Personal Portfolio Website *(React, Vite, Tailwind CSS)*

- Designed and built a personal portfolio website from scratch using React and Vite, deployed on Vercel with a custom domain, CI/CD, and SSL.
- Configured SEO fundamentals and built fully responsive, mobile layouts across every section.

Technical Skills

- **Programming:** Java, Scala, SQL, Python, HTML/CSS, Assembly (AT&T x-86)
- **Software Engineering:** Spring Boot, JUnit, JavaFX, React, Vite, Tailwind CSS, REST APIs
- **Big Data and Distributed Systems:** Apache Spark, Apache Flink
- **Cloud and AI:** Azure DevOps, Azure AI Foundry, Azure AI Search, Blob Storage, Copilot Studio
- **Databases:** PostgreSQL, MySQL
- **Tools:** Git, Unix/Linux, JSON, Markdown and CommonMark

Languages and Interests

English - Native | Dutch - Proficient (C1) | Gujarati - Native
Badminton Association - DSBA-USSR (Delft Student Badminton Association)
ICPC Algorithmic and Programming Competition (BENELUX Preliminaries)